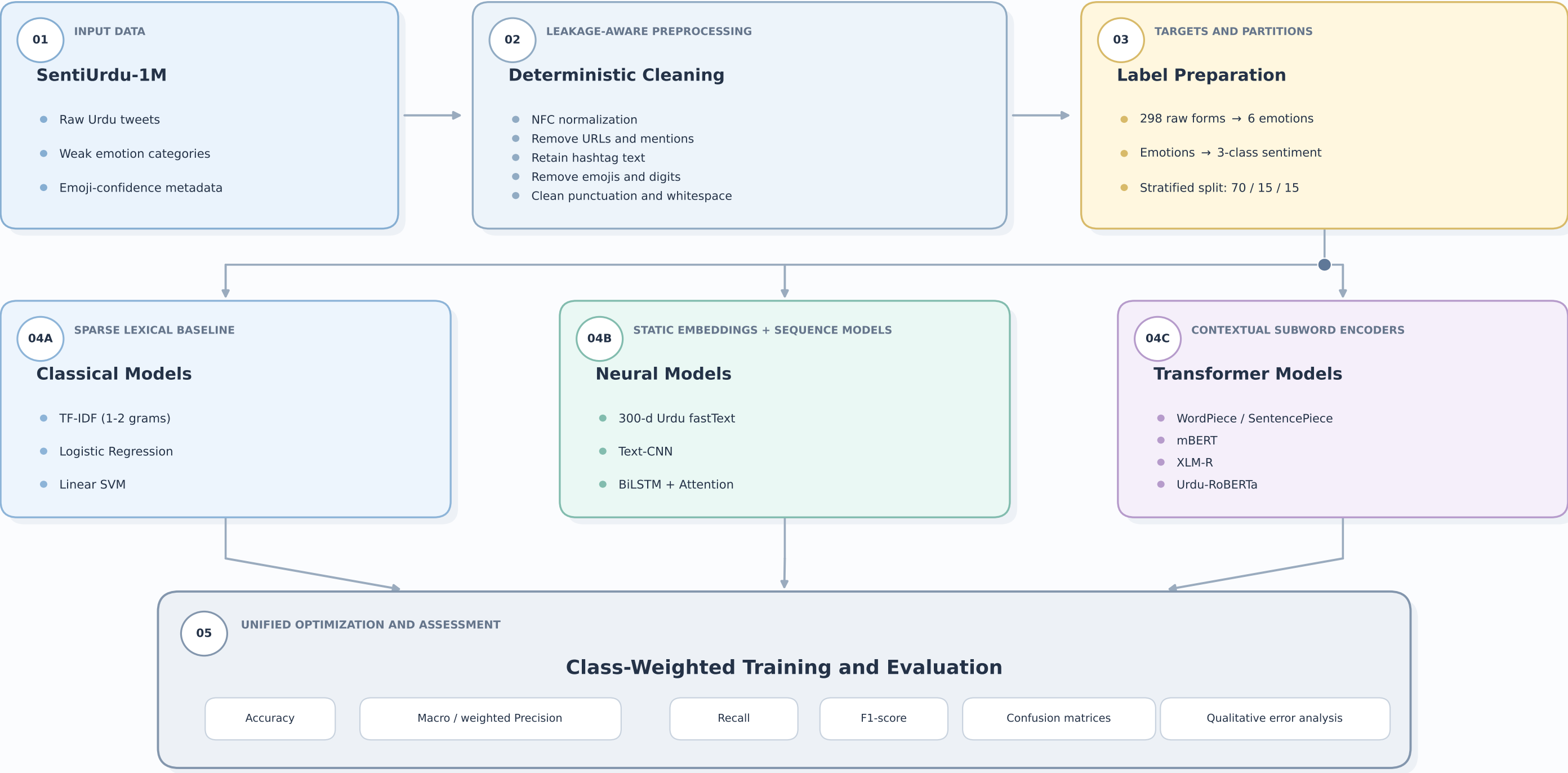


Leak-Free Urdu Sentiment and Emotion Classification Pipeline

Controlled data preparation, three model families, and unified class-sensitive evaluation



Leakage Prevention Note

Emojis are removed before feature extraction to prevent leakage from the weak-labeling heuristic.